GeoGenBench: Symbolically Verified Generation of Geometry Diagrams

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Abstract

Typically, geometry benchmarks ask models to read and interpret diagrams. In GeoGenBench, we ask the models to draw. Each of the 801 prompts describes a plane-geometry construction and comes with the checks a correct drawing has to pass. The model writes TikZ. We compile the code, read the coordinates back out, and run every check exactly, so no person and no vision model has to judge the drawing. We test six frontier models. Asking for a typed intermediate representation instead of raw TikZ helps every one of them, by 2 to 25 points, and helps the weakest the most. With that scaffold, the three best models land within a point of each other at prices that differ sevenfold. The grader has one known gap. Every check it runs is exact, but the checks do not always cover everything a prompt asks for. When two people graded 200 outputs by hand, the grader never failed a correct drawing, but it passed about one drawing in six that a person would not. We release the dataset, the templates, the grader, and the leaderboard. The remaining work is making each template's checks cover every requirement in its prompt.

1 Introduction

A language model asked to draw a geometry diagram will usually produce something that looks right. Looking right is not the same as being right. Geometry benchmarks have tested reading: here is a diagram, answer a question. Drawing has gone untested, because grading a drawing has been the hard part.

Take the prompt *"Draw an acute triangle EFG with angles $60°$ and $70°$ at E and F. Construct the altitude from G meeting EF at H."* A correct drawing has those two angles, a right angle at H, H on the segment EF and not beyond it, and no obtuse angle. In our pilot, one flagship model put H beyond the segment. Another drew an obtuse triangle. Both drawings looked fine.

Two ways to grade such a drawing exist today, and neither works. A person can do it, slowly and inconsistently. A vision model can say whether the picture looks like a right triangle, but not whether the angle is $90°$. We do a third thing. We compile the model's drawing back into coordinates and test each requirement as arithmetic. We call each requirement a *check*, and a drawing that meets every check a *pass*. A right angle is then a calculation, not an opinion.

The rest of the paper follows one question after another. Section 2 says how the prompts and their checks are made and how a drawing is graded. Section 3 asks whether the grader can be trusted, and tests it against adversarial drawings and against people. Section 4 grades six frontier models. Section 5 says what the numbers mean.

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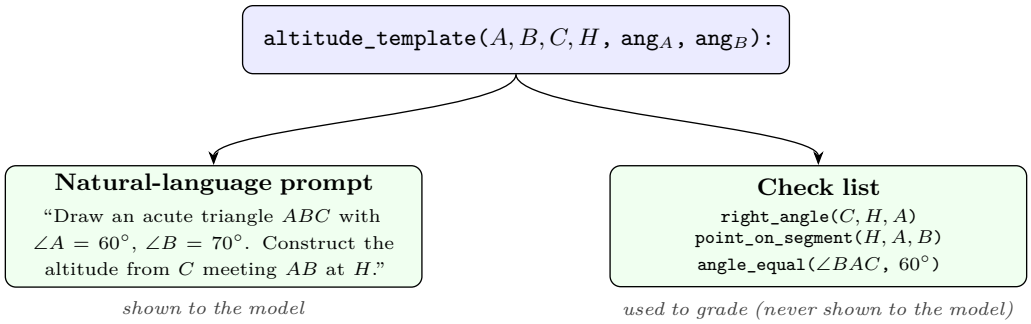


Figure 1: Procedural co-generation: one Python template emits both the prompt and the checks, so the two cannot disagree.

Prior work. Geometry3K [6], GeoQA [3], UniGeo [4], MathVista [7], GeoEval [12], and GeoBench [1] test diagram reading. T2I-CompBench++ [5] and GenExam [9] grade generated images with a model as judge. AlphaGeometry [8] and the Lean corpora machine-check proofs. GeoSDF [11] draws a diagram from symbolic constraints by optimizing a signed distance field. Geoint-R1 [10] checks auxiliary constructions in Lean 4. Both build the drawing themselves. GeoGenBench grades a drawing a language model built from a sentence. It could grade a drawing from GeoSDF the same way.

2 How a drawing is graded

Prompts and checks come from one template. A Python template writes both the sentence the model sees and the checks the grader runs (Figure 1). The altitude template returns the prompt above and the list `[right_angle(`A`,`H`,`B`), point_on_segment(`H`,`B`,`C`)]`. Because both come from the same source, a check can never ask for something the prompt did not. A reviewer worried about contamination can re-roll the template with a fresh seed.

From drawing to verdict. The model emits TikZ. We compile it twice: once to an image, once to SymPy geometry with coordinates. Fifteen checks, such as `right_angle`, `midpoint`, `parallel`, and `point_on_segment`, run on those coordinates in microseconds, with a small numeric tolerance. Two weaker checks on labels and marks count only toward a loose pass. Template authors say what must hold, not how to test it.

Scoring. The model sees only the prompt. An output passes if every check passes. It gets a soft pass if nothing fails but a check was skipped. Otherwise it fails, or it never rendered.

What is in the benchmark. 600 of the 801 prompts come from 30 templates, 200 per tier: **T1** is a single shape, **T2** adds one construction, **T3** combines several. The other 201 were extracted by a language model from a state K–12 curriculum [2]. We report the two sources separately and treat the templated 600 as the anchor.

3 Can the grader be trusted

Do the checks pin the drawing down. A check list is complete if every drawing that passes it is the intended construction, up to position and scale. Of the 30 templates, 18 are that tight. The other 12 pin the named relations but leave some parameters free, as a teacher's instructions usually do.

Can the checks be fooled. We tried to write TikZ that passes each check while contradicting the prompt. Five checks can be fooled on their own. Templates pair checks, so we estimate under 5% of passes are exploitable.

Table 1: Headline leaderboard (600 prompts × 3 repeats, post-hardening). Strict = `pass`; loose adds `soft_pass`. * marginally partial coverage; † blocked at the API layer (Section 4).

Model	Strategy	N	Strict	Loose	$/prompt	Latency
GPT-5.5	structured	2,304*	**94.1%**	94.1%	$0.075	33.6 s
Claude Sonnet 4.6	structured	2,861	93.4%	93.4%	$0.088	19.0 s
Claude Opus 4.7	structured	2,292*	93.3%	93.3%	$0.532	23.9 s
GPT-5.5	raw_code	1,800	91.9%	95.3%	$0.058	14.8 s
Gemini 2.5 Pro	structured	1,140*	83.1%	83.1%	$0.050	75.7 s
Claude Sonnet 4.6	raw_code	1,800	81.2%	95.3%	$0.044	16.4 s
Claude Haiku 4.5	structured	2,232	75.3%	76.8%	$0.032	8.0 s
Claude Opus 4.7	raw_code	1,800	71.6%	95.3%	$0.262	13.8 s
Claude Haiku 4.5	raw_code	2,403	70.5%	76.0%	$0.019	9.5 s
Gemini 2.5 Pro	raw_code	1,800	58.4%	89.9%	$0.049	32.9 s
Gemini 2.5 Flash	raw_code	2,403	35.9%	42.2%	$0.012	22.1 s
Gemini 2.5 Flash	structured	0†	—	—	—	—

Hardening. Three grader fixes promoted 4.6% of pilot outputs from soft pass to pass and demoted none. They also halved the apparent gap between the two strategies in Section 4. Part of that gap had been the grader's, not the models'. Every number below is post-fix.

Does it agree with people. Two raters, a PhD mathematician and an engineer who has taught mathematics, judged 200 outputs blind to the grader's verdict. Grader and raters agree moderately ($\kappa = 0.455$), below the 0.70 we had pre-registered, while the raters agree well with each other. The disagreement runs one way. The raters overturned 18% of the grader's passes and none of its fails. Where do those overturned passes live? We split the 200 outputs by tier. On T1 and T3, the grader and the raters agree almost every time. On T2 they barely agree, at $\kappa = 0.16$. T2 is the tier that adds one construction, an altitude or a median, to a triangle whose angles the prompt states. The raters saw that a drawing did not have those angles and failed it. The grader never measured them, because the template wrote no check for them. Two hundred outputs spread over six models leave too few per model to correct each score on its own, so we report the tier instead. Section 5 says what that means for the leaderboard.

4 Results

Models and strategies. Six models from three vendors at a 20× price spread: Claude Opus 4.7, Claude Sonnet 4.6, Claude Haiku 4.5, GPT-5.5, Gemini 2.5 Pro, and Gemini 2.5 Flash. Two strategies. raw_code hands the model the prompt and a TikZ tutorial and takes what it writes. structured asks for typed JSON describing points, segments, circles, and checks, which our code turns into SymPy and then TikZ. The checks run before rendering, so the model gets up to three retries. Every model-by-strategy cell runs on the 600 templated prompts three times (Table 1).

The scaffold lifts every model. structured helps each model: GPT-5.5 by 2 points, Claude Haiku 4.5 by 5, Claude Sonnet 4.6 by 12, Claude Opus 4.7 by 22, and Gemini 2.5 Pro by 25. The weaker a model is at raw TikZ, the more it gains. The scaffold changes three things at once. The model writes a typed list instead of drawing code. Our code compiles that list. And the model may retry, up to three times, when its own checks fail. Raw TikZ gets none of these. We have not separated them, so the gain belongs to the scaffold as a whole, not to any one part of it.

The frontier converges. With the scaffold, GPT-5.5, Claude Sonnet 4.6, and Claude Opus 4.7 finish at 94%, 93%, and 93%, within one point of each other. Claude Opus 4.7 costs seven times what GPT-5.5 does. Without the scaffold the spread runs from 92% down to 36%, and Claude Opus 4.7 is the most dominated cell, 20 points behind GPT-5.5 at four times the cost.

Tiers separate the strategies, then the models. T1 is nearly solved by every Anthropic and OpenAI model under the scaffold. T2 is where the scaffold first matters: it lifts Claude Opus 4.7 by 43 points and Gemini 2.5 Pro by 54. T3 is where the frontier models separate: GPT-5.5 93%, Claude Sonnet 4.6 87%, Claude Opus 4.7 82%.

Wrong drawings, not broken code. Most failures are clean, rendered diagrams with the wrong geometry. The grader catches them. A person glancing at the picture often would not.

Portability. structured needs schema-constrained decoding. Our schema compiles to thousands of decoder states, and one vendor's API refuses it before the model writes a token. That is a gap in the API, not the model, and a warning for any benchmark built on a large schema.

5 What this means

Exact grading is practical. Every drawing in Table 1 was graded by a program, in microseconds, with the same verdict every time. A model judge would have cost roughly a thousand times more per drawing and could not have told a right angle from one that looks right.

The scaffold matters more than the model. The largest effect in the study is not which model you buy but whether you ask it for a typed representation before it draws. That representation lets the model check its own work before rendering. The most expensive model gains the most from it and, once it has it, no longer earns its price.

The grader errs in one direction only. Every drawing it fails is wrong. About one pass in six is not as right as it looks, almost always on T2, where the checks confirm the construction but not the numbers the prompt gave. That lenience applies to every model on the same prompts, so the rankings hold even where the absolute rates are a little high. Numeric checks for those templates are the planned fix.

Limitations. GeoGenBench tests whether a model can turn a sentence into a consistent plane drawing. It does not test proofs, solids, or teaching. A third of the templates accept any drawing that meets the stated constraints, not one canonical drawing. The headline numbers are templated numbers. On T1 and T2 the top three models are near the ceiling, and a test every model passes stops telling them apart. The curriculum prompts are far from the ceiling. We ran three model-and-strategy cells on them, and every score roughly halved. Claude Haiku 4.5 under raw TikZ fell from 82% to 36%. Thirty templates is a small set. Each is one Python function that writes its prompts and its checks together, so a new template is cheap to add. The next two steps are to run every model on the curriculum prompts and to write more templates.

Takeaway. Each requirement on a geometry drawing can be checked exactly, from its coordinates, with no judge. The remaining work is making sure each template's checks cover every requirement in the prompt. Graded this way, the scaffold matters more than the model. We release everything needed to run it.

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